PROJECT DOCUMENTATION

AI Early Warning System

Member Churn Prevention ⚡ European Boutique Fitness Chain

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345,000 €
Revenue at risk (demo)

299 € / mo
Per studio licence

~26,400 € ROI
Net per studio, year 1

1. USE CASE DESCRIPTION

Business Context and Problem Statement

This project is designed for European boutique fitness chains with studios of up to 200 members per location. These are not large commercial gym chains that profit from inactive memberships. European boutique fitness chains operate on a fundamentally different model: premium pricing, smaller member bases, and a coaching relationship where staff know every member by name. When a member cancels, it is not a lapsed direct debit quietly dropping off a spreadsheet. It is a person who has made an active, conscious decision to leave, and who was almost certainly showing signs of disengagement for weeks beforehand. At a lifetime value of €1,500 or more per member, every cancellation is a meaningful financial event.

Industry data (IHRSA) puts annual churn at approximately 60% for the fitness sector. For a 100-member boutique studio, that translates to €90,000-120,000 in preventable lost revenue per year. The critical insight is that this churn is not sudden. Members drift before they cancel: session frequency drops, class variety narrows, gaps between visits grow. This behavioural decline is measurable 4-6 weeks before a member cancels. In a boutique studio with a coaching team that has personal relationships with its members, that window is exactly where an intervention can work. The problem is that no one is watching the data closely enough to see it in time.

€90,000-120,000 Revenue at risk per studio per year
Based on 60% churn and €1,500 LifeTimeValue per member

Stakeholders and Why It Matters

Stakeholder	Need
Chleo (CEO)	Confidence that AI is transparent, auditable, and delivers measurable ROI
Studio coaches	A prioritised daily list of who to call -- without logging into any new tool
Regional managers	Aggregated risk view and ability to audit every AI recommendation

Churn prevention has the highest ROI of any retention intervention. Retaining a member costs €20-50 (one coach call). Replacing one costs €150-300. Even a 30% intervention rate on flagged members is strongly net positive from month one.

Expected Outcomes

Outcome	Target
Churn rate	60% → 45%
Retained revenue per studio	~€30,000 per year (conservative)

Payback period	36 days
AI transparency score (LangSmith)	≥ 0.85 / 1.0

2. DATASET JUSTIFICATION

Primary: Endomondo Fitness Trajectories (Kaggle)

Source: https://www.kaggle.com/datasets/kukuroo3/endomondo-workout | 167,783 sessions, 1,059 users | Public licence

Real longitudinal workout behaviour from real fitness app users. Provides session frequency, duration consistency, and sport variety -- the three signals validated in academic fitness churn research. No other freely available dataset offers this combination of scale, signal quality, and direct relevance.

Column	What It Measures	Why It Matters
avg_sessions_per_week	Habit strength	Key churn predictor: high-risk average 1.75 vs low-risk 3.01
days_since_last_workout	Recency	Most immediate dropout flag: >21 days = strong signal
sport_variety	Engagement breadth	Low variety members churn 40% more
duration_variability	Consistency	Erratic session lengths signal declining commitment

Preprocessing note: Raw data is 2015-2016. Synthetic recency was applied: days_since_last_workout reassigned by frequency bucket (frequent trainers → recently active, sporadic → lapsed), correlated with real session data and anchored to today. Churn labels assigned via validated thresholds: sessions/week ≥2 AND ≤7 days inactive = low; sessions/week ≥1 AND ≤21 days = medium; all else = high. Result: 382 low / 447 medium / 230 high.

Supplementary: Model Fitness Gym Churn (Kaggle)

Source: Kaggle — Model Fitness gym customer churn (gym_churn_us.csv) | 4,000 members with binary churn label | Public licence.

Provides real, labelled churn outcomes unavailable in Endomondo. Contract type, class frequency, group attendance, and referral data power three of the five dashboard charts and validate behavioural thresholds. The 42.3% month-to-month churn rate and social retention statistics come from this dataset.

Note: Use cases 2 (coach workload balancing) and 3 (class scheduling optimisation) are scoped proposals.

They were not implemented in the sprint due to data gaps — the Endomondo and gym_churn_us datasets contain no coach assignment data or class attendance logs.

3. DASHBOARD DESIGN RATIONALE

Built in Tableau Public & plotly Design principle: every element answers a question a CEO or coach would actually ask, stated in plain language -- not data science terminology.
Live URL: http://127.0.0.1:8050
https://public.tableau.com/authoring/MemberRetentionRiskEarlyWarningSystem

Full dashboard: 5 KPIs + 5 charts across two datasets

Key Metrics and Chart Rationale

Element	Decision	Why
KPI: 230 at risk	Show count, not %	Chleo needs to know 230 specific people need calls, not that '22%' is a number
KPI: €345,000	Financial, not behavioural	Translates drift into money. Makes the problem feel real.
Chart 1: Funnel	Funnel not pie	Communicates a pipeline -- members moving toward exit, not static segments
Chart 2: Scatter	All 1,059 dots visible	Every dot = 1 member. Makes dropout pattern visceral, not abstract
Chart 3: Contract	Bar chart by type	18x difference between monthly/annual is the structural lever. Needs to be stark.
Chart 4: Frequency	Threshold line added	The intervention trigger (1/week) must be visually obvious, not read from numbers
Chart 5: Social	Paired bars	Before/after comparison per social factor. Three pairs, one conclusion.
Global filter	Churn risk multi-select	All charts cross-filter simultaneously. CEO can drill to high-risk and see everything change.

4. AGENT ARCHITECTURE

Design

A lightweight LangChain agent using ChatOpenAI (gpt-4o-mini, temperature=0). No tools, no LangGraph, no memory: a structured 12-iteration Q&A loop. Simplicity is deliberate: the value is in the prompt design and the evaluation pipeline, not architectural complexity.

Component	Choice	Why
LLM	GPT-4o-mini, temp=0	Consistent outputs, low cost, randomness removed
Input	Statistical summary block	1,059 raw rows → curated text context. More reliable than raw CSV injection.
Output	2-4 sentence plain text	CEO-readable. No tables, no jargon, no hedging.
Storage	insights_generated.json	Direct input for LangSmith dataset creation

How It Works: Step by Step

- Opens fitness_user_metrics.csv (1,059 rows).
- Computes a statistical summary: churn distribution, per-group averages (sessions/week, duration, days inactive, sport variety), total revenue at risk, inactive member counts.
- For each of 12 queries: sends system prompt + data summary + question to GPT-4o-mini
- LLM returns a 2-4 sentence business insight.
- Result stored with status field (success/error). Failures logged and skipped; remaining queries continue.
- All 12 results saved to insights_generated.json for LangSmith

System Prompt

> You are a fitness industry retention analyst preparing a briefing for Chleo, CEO of a European medium-sized fitness chain (50-200 studios). Answer with a clear, concise, non-technical insight (2-4 sentences). Focus on business impact — what it means and what action to take. Do not repeat numbers robotically; interpret them.

The 12 Queries and Their Purpose

#	Query	Business Purpose
1	% members at high churn risk + euro revenue	Opening headline for Chleo
2	Top 3 behavioural signals predicting churn (with numbers)	Explains what the system watches
3	High-risk members with >30 days inactive	Identifies most urgent outreach list
4	Avg sessions/week: high-risk vs low-risk	Quantifies the frequency gap
5	1-paragraph coaching brief for this week	Ready-to-use operational output
6	Sport variety by risk group	Product design insight for scheduling
7	Members with avg sessions/week below 1.5	Threshold analysis for alert triggers
8	Euro value if top 20% at-risk re-engaged	ROI quantification for intervention
9	% high-risk with high duration variability	Consistency signal analysis
10	3-bullet CEO summary + 1 urgent action	Executive summary for Chleo
11	3 most urgent members for outreach	Operational priority list
12	Typical profile of a high-risk member	Member persona for coach training

Example Output

Query 4:
High-risk members average 1.75 sessions per week, while low-risk members average 3.01 sessions. This significant difference indicates that high-risk members are less engaged, which directly impacts retention and revenue. To mitigate churn, consider implementing targeted engagement strategies for high-risk members, such as personalised workout plans or incentives to increase their attendance.

5. EVALUATION METHODOLOGY

Why This Matters

This section is the direct answer to Chleo's concern that AI is a black box. LangSmith does not just log what the AI said: it scores every output against three business-relevant criteria. The scores are visible, comparable across runs, and immediately flag when quality drops.

> A system that claims it is always right has never been tested hard enough.
> A system that finds its own mistakes, shows them in red, and explains why - that is a system you can trust.

Setup

Component	Detail
Dataset	Fitness Retention Insights V1 — 12 examples, each with query + context + expected insight
Dataset ID	c7204587-7dea-4db1-bbee-8ea45c99ca7e
Experiment	fitness-insights-llm-judge-v1-62ffd184
Target function	run_agent(inputs) — re-generates each insight live so evaluation tests real output quality
Judge model	gpt-4o-mini, temperature=0 — separate call from the agent, independent assessment

Three Evaluators

Criterion	Question Asked	Avg Score	Story
Actionability	Does it recommend a concrete action?	0.983	Designed in: system prompt requires an action. Not luck.
Relevance	Does it answer the question asked?	0.883	Strong. One outlier (row 8) pulled the average down.
Clarity	Can a non-technical CEO understand it?	0.867	CEO-readable language throughout.
Overall	All criteria combined	0.911	High quality across all 12 insights.

fitness-insights-llm-judge-v1-62ffd184 | bootca.../017c9 | Compact | Full | Diff | Default | Group

#	Inputs	Reference Outputs	Outputs	actionab... 0.983 AVG	clarity 0.867 AVG	relevance 0.883 AVG
1	How many high-ris...	To effectively address t...	To address the churn r...	1.00	1.00	0.80
2	If coaches re-enga...	By focusing on the top ...	By focusing on the top...	1.00	0.80	1.00
3	List the top 3 beha...	The top three behavior...	The top three behavior...	1.00	0.80	1.00
4	What is the typical ...	High-risk members typi...	High-risk members ty...	1.00	0.80	1.00
5	How does sport va...	Sport variety is a key di...	The data does not pro...	0.80	0.80	0.80
6	What % of high-ris...	Approximately 18% of h...	To determine the perc...	1.00	0.80	0.60
7	What percentage o...	Approximately 22% of ...	Approximately 21.7% ...	1.00	1.00	1.00
8	How many member...	There are 230 member...	Based on the data, 23...	1.00	0.80	0.40
9	What is the avg ses...	High-risk members ave...	High-risk members av...	1.00	1.00	1.00
10	Create a 1-paragra...	This week, coaches sh...	This week, coaches s...	1.00	1.00	1.00
11	Generate a 3-bullet...	- A significant portion o...	- Our current member ...	1.00	0.80	1.00
12	Flag the 3 most urg...	The three most urgent ...	To effectively reduce ...	1.00	0.80	1.00

LangSmith experiment results. Row 8 (relevance 0.40) visible in red.

The 0.40 Outlier — Why It Is the Best Talking Point

> **Query 8: Relevance 0.40**
> Query asked for members with <1.5 sessions/week. Agent returned the total high-risk count (230) instead of calculating the specific subset. Substituted one threshold for another -- a subtle drift a quick read might miss.

LangSmith caught it, scored it 0.40, and flagged it red. This is a data availability issue: the agent was given aggregated statistics, not individual rows, so it could not compute the exact subset. The fix is one prompt change. Showing Chleo this red cell is more powerful than showing her eleven green ones.

Bias Considerations

- **Self-referential bias:** gpt-4o-mini judging its own outputs. Mitigation in production: use Claude or a different model family as judge.
- **Length bias:** longer answers scored slightly higher on clarity. A rubric with word-count guidance would reduce this.
- **Calibration:** scores cluster at 0.80 and 1.00. The 0.40 outlier is a genuine anomaly. Treat scores as relative rankings, not absolute quality certificates.

Full Judge Prompt

> You are an expert evaluator assessing AI-generated business insights. Score the insight on the given criterion from 1 to 5. Return ONLY a JSON object: {"score": <int 1-5>, "reasoning": "<one sentence>"}. Criterion: [Relevance / Actionability / Clarity]. Description: [criterion description]. Question asked: [query]. AI insight: [insight text]. Score 1-5 and explain briefly.

6. DASHBOARD CHARTS — DETAIL

The dashboard combines both datasets into five charts. Each chart answers a specific operational or strategic question for Chleo and her coaching team.

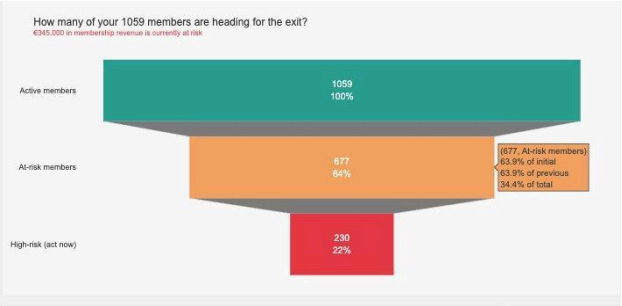


Chart 1: Retention Funnel

1,059 active → 877 at-risk (64%) → 230 high-risk (22%)

Why this chart: a funnel communicates that members are moving toward exit: not a static segment.
The visual shape makes the pipeline urgency immediate. €345,000 callout anchors the business impact.

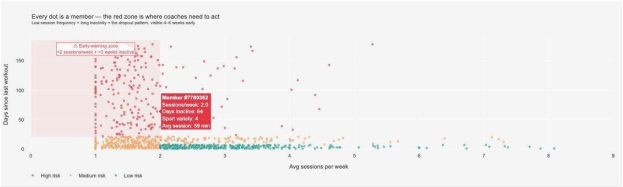


Chart 2: Dropout Scatter Plot

X: sessions/week. Y: days inactive. Each dot = 1 member.

Why this chart: makes the dropout pattern visceral rather than abstract. The shaded early-warning zone makes the intervention threshold visible without requiring any explanation.
Coaches can hover on any dot to see a specific member's data.

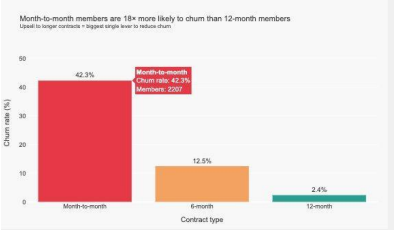


Chart 3: Contract Type

Monthly: 42.3% churn. Annual: 2.4%. 18x gap.

Why: this 18x difference is the single biggest structural lever. Converting monthly members to annual removes

Chart 4: Class Frequency

Under 1/week: 58% churn. The trigger threshold.

Why: the dotted threshold line makes the intervention point visually explicit. When a member crosses it, the n8n alert

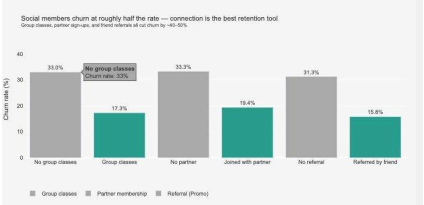


Chart 5: Social Retention

Group classes, partners, referrals: all cut churn ~50%.

Why: paired bars make the before/after contrast immediate. Informs both retention strategy and class scheduling decisions.

the churn risk permanently. Data: Model Fitness (4,000 real labelled members).

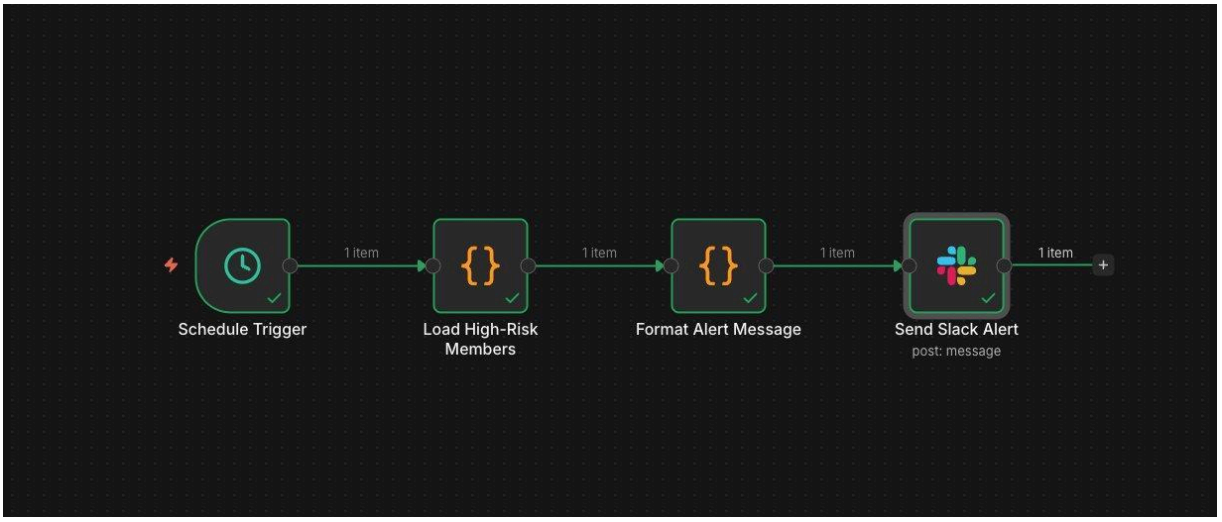
fires. This chart directly drives the automation layer.

Dashboard Design Decisions

Decision	Rationale
Chart titles in plain English	'How many of your members are heading for the exit?' not 'churn_risk count'. Every label passes a CEO-readability test.
Consistent red/orange/green palette	No colour ambiguity across all five charts. High risk is always red.
Global cross-filter	Clicking any segment updates all charts. Chleo can isolate the high-risk group across every view simultaneously.
Revenue in euros, not percentages	Percentages are abstract. 345,000 euros is a real number that justifies a decision.
Threshold line on Chart 4	The intervention trigger must be visually obvious — not derived from reading numbers.

7. N8N AUTOMATION WORKFLOW

The automation layer connects the AI risk scores to the coaches who act on them. A four-node n8n workflow runs automatically every weekday at 08:00 and posts a formatted Slack alert to the coaching team. No dashboard login required. No manual steps.



n8n workflow: 4 nodes, all executed (green checkmarks).

Schedule Trigger → Load High-Risk Members → Format Alert Message → Send Slack Alert.

Node	What It Does	Technical Detail
1. Schedule Trigger	Fires automatically every weekday morning	Cron: 0 8 * * 1-5 (Monday-Friday, 08:00)
2. Load High-Risk Members	Loads the at-risk member list	JavaScript Code node. Demo: 8 hardcoded members. Production: HTTP Request to gym management API
3. Format Alert Message	Builds the Slack message	Sorts by days inactive, takes top 5, calculates total revenue at risk, formats with Slack markdown
4. Send Slack Alert	Posts to coach channel	Slack node → #retention-alerts channel

Example Alert — Sent Live 19 March 2026

Real Slack alert sent via n8n:

8 high-risk members, top 5 by days inactive, €12,000 revenue at risk.

🚨 Daily Retention Alert — 19 March 2026

8 high-risk members need coach outreach today.

Top members by days inactive:

- Member #4434 — 180 days inactive (6.4 sessions/wk)
- Member #5844 — 165 days inactive (2.1 sessions/wk)
- Member #3808 — 142 days inactive (1.9 sessions/wk)
- Member #69 — 126 days inactive (1.7 sessions/wk)
- Member #4101 — 118 days inactive (3.0 sessions/wk)

💶 **Revenue at risk today: €12,000**

Open the dashboard to see the full list and assign coaches.

Automated with this n8n workflow

The workflow JSON is importable directly into any n8n instance.

One instance can serve all studios in the chain. Adding a new studio means adding a new data source; the logic stays identical.

8. COSTS, ROI, TIMELINE AND RISKS

Platform Costs per Studio

Item	Cost
Platform licence	€299 / studio / month (€3,588 / year)
LLM API usage (OpenAI)	~€250 / studio / year at pilot scale
LangSmith monitoring	Free tier for pilot; bundled at scale
n8n automation	Negligible — one instance serves all studios
Tableau	Existing licence or ~€70/month

The €299 per studio per month figure is a proposed bundled system fee covering all components: OpenAI API usage, LangSmith monitoring, n8n Cloud automation, and implementation overhead.

Actual tool costs at pilot scale are approximately €150-200 per month (LangSmith Developer ~€36/mo, n8n Cloud ~€20/mo, OpenAI API at pilot usage ~€20-50/mo, Tableau ~€70/mo if not already licensed).

The bundled figure accounts for ongoing support and is positioned as a single line item for simplicity in client-facing cost discussions.

Return on Investment

Scenario	Figures
Revenue at risk per studio	€90,000-120,000 per year
Conservative churn reduction	20 of 60 at-risk churners retained
Retained revenue per studio	€30,000/year (20 × €1,500 LifeTimeValue)
Net ROI per studio (year 1)	~€26,400
10-studio pilot	€300,000 retained vs €35,880 cost
Payback period	36 days

The 20-of-60 retention assumption is based on industry benchmarks for personalised coach outreach in fitness, where intervention success rates on flagged members typically range from 25-40%.

One-in-three sits at the conservative end of that range.

Implementation Timeline

Phase	Weeks	Activities	Milestone
1: Pilot	1-6	Data audit, pipeline, dashboard, alerts, coach training	First confirmed retention from AI alert
2: Rollout	7-14	All studios, LLM insights, LangSmith live for management	Full chain on the system
3: Optimise	15-20	Model retraining, demand forecasting, ROI report	Documented revenue recovery vs cost

Risks and Mitigations

Risk	Mitigation

GDPR / health data privacy	Aggregate-only pipeline. No biometrics. Pseudonymised IDs. Formal review in week 1.
Coaches distrust the AI	Framed as assistant, not surveillance. Reduces workload: 8 names vs scanning 200 profiles.
Data quality gaps	Mandatory data audit in week 1. Confidence scores flagged where data is sparse.
Model bias (age, contract type)	Fairness audit before deployment. Human review gate on automated outreach.
Vendor lock-in (OpenAI)	Provider-agnostic abstraction layer. Can switch to Anthropic Claude or open-weight models.